

1. Administrative & Study Identification

Full Study Title: A Clinical Study of Letemovir (MK-8228) in Children and Adolescents Who Receive a Kidney Transplant (KT) (MK-8228-077)
Official Title: A Phase 1, Open-label, Single-arm Study to Evaluate the Pharmacokinetics and Safety of Letemovir in Pediatric Kidney Transplant (KT) Recipients Less Than 18 Years of Age and Weighing Less Than 40 Kilograms **Short Title/Acronym:** MK-8228-077
Protocol Number: MK-8228-077
NCT Number: NCT07199465
Posting ID: 991037922356602273 **Sponsor/Company Name:** Merck
Investigational Product: Letemovir (MK-8228 / PREVYMIS™)
ATC Code: J05AX18 **Phase of Development:** PHASE1
Trial Status: RECRUITING **Disease Area / Preferred UMLS Name:** Cytomegalovirus Prophylaxis **SNOMED ID:** 1371348001 **Medical Monitor:** Merck Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (specifically, Area Under the Curve from time 0 to 24 hours) of letemovir, as well as to assess the overall safety and tolerability in children and adolescents who have received a kidney transplant. **Secondary Objectives:** To evaluate the percentage of participants experiencing adverse events (AEs) and the percentage of participants discontinuing the study treatment due to AEs.

2.2 Background & Rationale To address the critical need for safe and effective cytomegalovirus (CMV) prophylaxis in the pediatric and adolescent solid organ transplant population. Researchers are looking for a way to prevent CMV in children and adolescents who receive a kidney transplant and weigh less than 40 kg, establishing proper dosing and identifying what happens to letemovir in the body over time, as well as learning about its safety and tolerability.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional
Control Type: Single-arm (Pharmacokinetic evaluation)
Blinding: Open-label
Randomization: N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 40 participants
Number of Sites: ~11 locations
Estimated Study Start: 24-Apr-2026
Estimated Primary Completion: 14-Apr-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Is a recipient of a primary or secondary allograft kidney. 2. Is at least 4 weeks posttransplant and not more than 52 weeks posttransplant at the time of enrollment (Day 1) and is being managed per local standard of care. 3. Has stable kidney function posttransplant. 4. Has undetectable CMV deoxyribonucleic acid (DNA) from a plasma or whole blood sample collected within 14 days prior to enrollment. 5. Must be able to take (as assessed by the investigator) letermovir tablets or oral pellets by mouth, or via gastrostomy or nasogastric tube (oral pellets only). 6. Does not have a condition that may interfere with the absorption of oral medication (e.g., vomiting, diarrhea, or a malabsorptive condition) from the day of enrollment (Day 1) until the intensive pharmacokinetics (IPK) sampling is completed. 7. Weighs ≥ 2.5 kg and < 40 kg at enrollment (Day 1).

4.2 Exclusion Criteria 1. Has CMV disease or suspected CMV disease between screening and enrollment. 2. Is on dialysis or plasmapheresis at the time of enrollment. 3. Has evidence of CMV viremia at any time from screening until the time of enrollment. 4. Has Child-Pugh B or C hepatic insufficiency within 14 days before enrollment. 5. Is a multi-organ transplant recipient (e.g., kidney-pancreas). 6. Has any uncontrolled infection on the day of enrollment. 7. Requires mechanical ventilation, or is hemodynamically unstable, at the time of enrollment. 8. Has received or is receiving protocol-specified prohibited medications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered daily.
Route of Administration: Orally (as tablets or oral pellets) or via gastrostomy or nasogastric tube (oral pellets only). **Duration of Treatment:** 7 consecutive days.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care post-transplant immunosuppressants.
Prohibited: Protocol-specified prohibited medications that may interfere with the absorption of oral medication or interact with letermovir.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, CMV DNA testing
Baseline	Day 1	Enrollment, First Dose, Vital Signs
Interim	Days 2 to 6	Daily medication administration, Safety Monitoring
End of Treatment	Day 7	Final Dose, Intensive Pharmacokinetics (IPK) blood sampling (Predose and up to 24 hours post-dose)

7. Outcome Measures (Endpoints)

7.1 Primary Outcome 1. Area Under the Curve From Time 0 to 24 Hours (AUC₀₋₂₄): Blood samples will be collected at multiple time points to estimate AUC₀₋₂₄.

7.2 Secondary Outcomes 1. Percentage of Participants Who Experience an Adverse Event (AE). 2. Percentage of Participants Who Discontinue Study Intervention Due to an AE.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Any unfavorable and unintended sign (including abnormal laboratory findings), symptom, or disease temporally associated with the use of letermovir. **Serious Adverse Events (SAEs):** Evaluated throughout the study period, requiring reporting within 24 hours. **Data Monitoring Committee (DMC):** Internal safety review committee established by Merck.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Set (all subjects with evaluable intensive PK sampling data); Safety Set (all subjects receiving at least one dose). **Sample Size Justification:** Sized to provide adequate precision for the estimation of letermovir AUC₀₋₂₄ in the pediatric weight tiers with a targeted enrollment of 40 participants. **Handling of Missing Data:** Standard non-compartmental PK analysis imputation; missing safety data will not be imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Onsite and remote monitoring to ensure compliance with the intensive PK sampling protocol. **Audit Trail:** Maintained through validated Electronic Data Capture (EDC) systems.

11. Study Identifiers & Governance

Primary ID: MK-8228-077
Registry Number: NCT07199465
Posting ID: 991037922356602273 **Sponsor/Lead Organization:** Merck
Study Title: A Clinical Study of Letemovir in Children and Adolescents Who Receive a Kidney Transplant
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12. Clinical Framework (Prophylaxis Specific)

Primary Condition: Cytomegalovirus (CMV) Prophylaxis
Diagnosis Threshold: Undetectable CMV DNA post-transplant
Medical Methodology: Antiviral prophylaxis for solid organ transplant (Kidney)
Optimization Target: Target PK exposure (AUC0-24) profiling for safe pediatric dosing

13. Study Procedures & Methodology

Management Pathway: Intensive Pharmacokinetics (IPK) evaluation framework
Education Protocol (HCP): Training on proper preparation and administration of oral pellets via gastrostomy or nasogastric tube
Education Protocol (Patient): Adherence to 7-day consecutive dosing schedule
Quality Audit Mechanism: Medication accountability and strict IPK timeline monitoring

14. Observation & Follow-Up Schedule

Total Duration: 7 days of treatment plus safety follow-up
Onsite Visit Cadence: Pre-screening, Baseline (Day 1), and Intensive Sampling (Day 7)
Remote Monitoring Frequency: Standard post-transplant clinical care cadence
Checklist Review Interval: Daily during the 7-day IPK period

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |